2-Minute Video Presentation Guide & Script

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Assessment: Algions AI Labs

Project Domain: Hospital Outpatient Operations Decision Support • **Target Duration:** Exactly 2 Minutes (120 Seconds)

1. High-Level Project Overview (What You Need to Know)

Component	High-Level Overview (What to Keep in Mind)
What is this project?	An AI decision-support tool for hospital receptionists and clinic operations. It predicts if a patient is likely to miss a scheduled outpatient appointment so staff can send targeted reminders or manage backup schedule buffers.
Where did the data come from?	A de-identified synthetic dataset of **15,000 appointment records** modeled on the standard public **Kaggle Medical Appointment No-Shows** benchmark. Strictly compliant with HIPAA: **zero sensitive clinical data** (no diagnoses, medical histories, or medications). Uses only booking-time variables (lead time, age, distance, SMS sent, and past attendance history).
What ML models were used?	Evaluated 4 algorithms: **Baseline Logistic Regression** (unweighted), **Balanced Logistic Regression** (class-weighted), **Random Forest**, and **Histogram Gradient Boosting**. Balanced Logistic Regression was chosen as the primary model for its clean probability calibration, zero latency, and explainability.
Key Breakthrough & Business Cost	Because an empty doctor slot costs **~$180** while a reminder call costs only **~$15**, missing a no-show is 12x worse! By adjusting our decision threshold from 0.50 to **0.35**, our model captures **97.7% of all no-shows**, cutting wasted clinic capacity costs by **76%** ($43.01 down to $10.18 per appointment).
Governance & Ethics	This is purely a **decision-support tool** for administrative reminders. It is **never used to deny care**, alter treatments, or triage clinical urgency. High risk triggers proactive support (like transport help or rescheduling).

2. Quick Setup Before You Press Record

Step 1: Start Server	Open Terminal in project folder and run: python3 app/app.py (Server runs on http://127.0.0.1:5001).
Step 2: Prepare Browser	Open http://127.0.0.1:5001 in Chrome or Safari. Ensure your name **Rohit Singh Negi** and **Reg No: 12311143** are clearly visible in the top navbar.
Step 3: Recording Tool	Use **QuickTime Player** (File → New Screen Recording), **Loom**, or **OBS**. Select full screen or browser window with clear microphone audio.

3. The Exact 2-Minute (120-Second) Spoken Script & Screen Actions

Speak at a natural, steady pace. You don't have to memorize every single word—just follow the bullet points!

Time	Screen Action (What to Show)	Exact Words to Speak (What to Say)
0:00 - 0:25 (25 sec)	**Website Home Screen** Show browser at http://127.0.0.1:5001 with navbar displaying your name and Reg No.	*"Hello everyone. My name is **Rohit Singh Negi**, registration number **12311143**. Today I am presenting my project: **Hospital Appointment No-Show Prediction**.* *In outpatient hospital operations, when a patient misses an appointment without notice, doctors sit idle and valuable clinic time is wasted. Our goal is to build an operational decision-support tool that predicts no-show risks at the time of booking, helping clinic staff direct phone reminders and organize scheduling buffers."*
0:25 - 0:50 (25 sec)	**Model Analytics Tab** Click the *'Model Analytics & Validation'* tab. Hover over Lead Time and Age plots.	*"For the dataset, respecting healthcare privacy and HIPAA standards, we used a de-identified outpatient dataset of 15,000 records modeled after the public Kaggle Medical Appointment benchmark. It strictly excludes sensitive clinical diagnoses.* *We engineered booking-time features like booking lead time, patient age, distance from clinic, SMS reminder status, and prior visit history. We used a strict **chronological split** to ensure zero data leakage from future visits into training."*
0:50 - 1:15 (25 sec)	**Cost Curve & Metrics** Point to the *'Cost Optimization Curve'* and metric cards (ROC-AUC 0.698, Recall 97.7%).	*"To handle the 4:1 class imbalance, we trained multiple models: baseline Logistic Regression, Balanced Random Forest, HistGradientBoosting, and Balanced Logistic Regression.* *Because an empty doctor slot costs about $180 while a quick phone call costs only $15, missing a no-show is 12 times more expensive! By optimizing our decision threshold to **0.35**, our model catches **97.7% of all no-shows**, cutting wasted clinic costs by **76%**."*
1:15 - 1:45 (30 sec)	**Live Demo: Single & Batch** 1. Switch to *'Single Patient'* tab. 2. Click **'Load High Risk Sample'** → **'Evaluate'**. 3. Switch to *'Batch Schedule'* tab → Click **'Run Sample Clinic Day'**.	*"Now let's see the live web dashboard. If I load a high-risk sample—a 23-year-old booking 28 days ahead with past missed visits—the model outputs a **68% risk probability**, explaining the top risk drivers and recommending a direct telephone call 48 hours prior.* *In the **Batch Scoring** tab, clinic staff can evaluate an entire day's appointment roster in one click to prioritize who needs reminders."*
1:45 - 2:00 (15 sec)	**Governance Tab & Closing** Click *'Governance & Limitations'* tab, then show VS Code / terminal briefly.	*"Finally, this system is strictly a decision-support aid—it is never used to deny care or triage medical urgency. The project includes 12 automated unit tests passing at 100% and is published on GitHub. Thank you!"*

4. Quick 3-Point Mental Cheat Sheet (If You Get Stuck)

1. Problem: Hospital no-shows waste physician slots ($180 loss). A reminder phone call is cheap ($15).

2. Data & Model: 15k de-identified records, no clinical data. Balanced Logistic Regression catches 97.7% of no-shows.

3. App Demo: Live Flask dashboard with single evaluation, risk gauge, local explainability, and batch roster scoring.